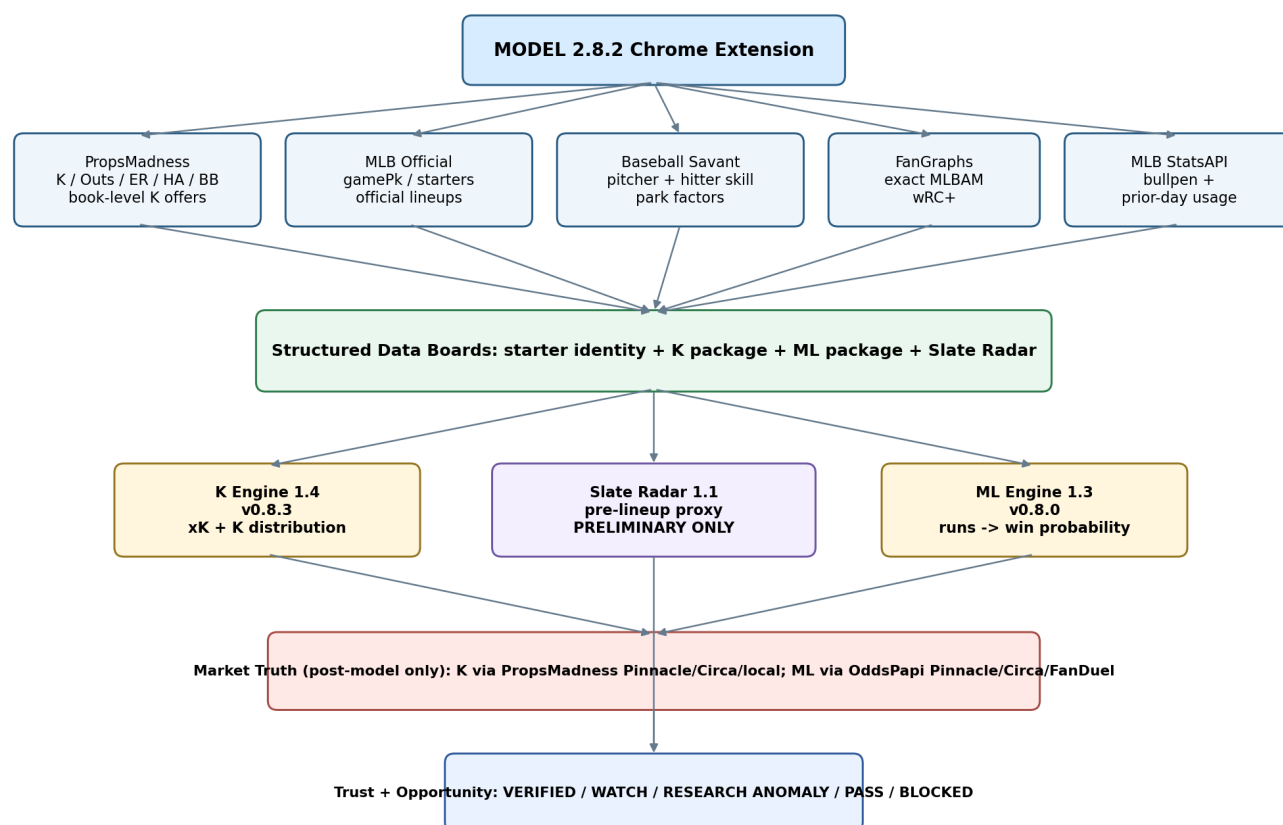


MODEL

MLB MODEL 2.8.2

User Manual, Technical Reference, and Project Checkpoint

Current as of September 8, 2026



Purpose. This document is the durable explanation of how MODEL works today, how to operate it, how its K and moneyline probabilities are calculated, what the local service does, what the Trust layer means, and exactly where the project stands for future development.

This is a probabilistic research tool. A model edge is not a guarantee of profit. The architecture is intentionally designed to separate prediction from market price and to fail closed when identity, workload, freshness, or data quality is insufficient.

1. Current State at a Glance

Component	Current version / policy
Chrome extension	MODEL 2.8.2 - Opportunity Integrity
Local market service	2.3.5 - MODEL Local Market Service on 127.0.0.1:8765
Structured K engine	1.4 - mlb-k-v0.8.3-sample-shrinkage-workload-2026-09-07
Moneyline engine	1.3 - mlb-moneyline-v0.8.0-offense-strength-2026-09-06
Slate Radar	1.1 - PRELINEUP_ACTIVE_ROSTER_PROXY - preliminary only
K market truth	PropsMadness book-level offers; Pinnacle/Circa same-line no-vig when available
ML market truth	OddsPapi game markets; Pinnacle + Circa sharp consensus; FanDuel executable price
OddsPapi quota policy	250 requests/month; normal MLB full market refresh ~3 requests; K uses 0
Legacy runtime	PMV1Engine / Hydrate / Batch runtime physically removed from active tree
Calibration status	K prospective after input migration; ML production input migration - forward validation still required
Next major branch	NFL target: MODEL 2.9.0; not yet a production release

Most important operating principle. BUILD frequently for free; spend game-market requests sparingly; allow Trust to decide whether a model signal is usable. Radar is scouting, not betting advice.

What is already live-verified?

- Structured ML production probability is independent of OddsPapi and has been live-verified end-to-end.
- OddsPapi 2.3.5 independently joins bookmaker-specific fixture IDs to MLB Official and merges them under canonical gamePk.
- A full 15-game MLB market refresh on September 8 produced 15 sharp-ready rows, 15 two-sharp consensuses, 15 multi-book merges, and 15 canonical games using 3 requests total.
- K v0.8.3 is live with small-sample shrinkage, workload states, confidence downgrades, and zero target-K-line influence on expected K.
- Opportunity Integrity is live: VERIFIED outranks WATCH; WATCH is explicitly not verified; anomalies are quarantined as research; missing prices stay null rather than becoming fake zeros.

2. Contents

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Checkpoint recovery. If project context is ever lost, upload this PDF and instruct ChatGPT: "Read Section 20 - Project Checkpoint and continue the MLB MODEL project from that state."

3. Philosophy: What MODEL Is Trying to Do

MODEL is built around a strict separation between baseball prediction and sportsbook price. The model first asks what should happen based on baseball inputs. Only after that probability exists does the market enter the calculation.

```
BASEBALL DATA
-> independent model probability
-> sharp market probability
-> executable sportsbook price
-> edge + EV
-> Trust gate
```

Why can an independent model disagree with a bookmaker?

A sportsbook price is a powerful information source, but it is not a literal statement of objective truth. Prices contain margin, update timing, inventory/risk considerations, correlated public information, and the bookmaker's own model. MODEL can disagree because it uses a different structured information set or interprets workload, lineup, starter, bullpen, and player-skill information differently.

- The edge can be real: MODEL may incorporate lineup-specific or workload-specific information more effectively or earlier.
- The edge can be temporary: the market may move toward the same information later.
- The edge can be false: bad data, a missed injury, role uncertainty, a small sample, or model misspecification can create a convincing-looking but wrong probability.

For that reason, MODEL does not assume that a larger edge is automatically better. A very large disagreement can be treated as an ANOMALY and sent to research rather than promoted as a bet.

The three probabilities that must stay separate

Concept	Meaning
Model probability	MODEL's independent estimate before sportsbook price is considered.
Sharp market probability	A no-vig benchmark from sharp books - currently Pinnacle/Circa when available.
Executable break-even probability	The probability required to profit at the price actually available to the user, usually FanDuel/local price.

Core market math

```
American odds -> implied probability:
negative odds: p = (-odds) / ((-odds) + 100)
positive odds: p = 100 / (odds + 100)

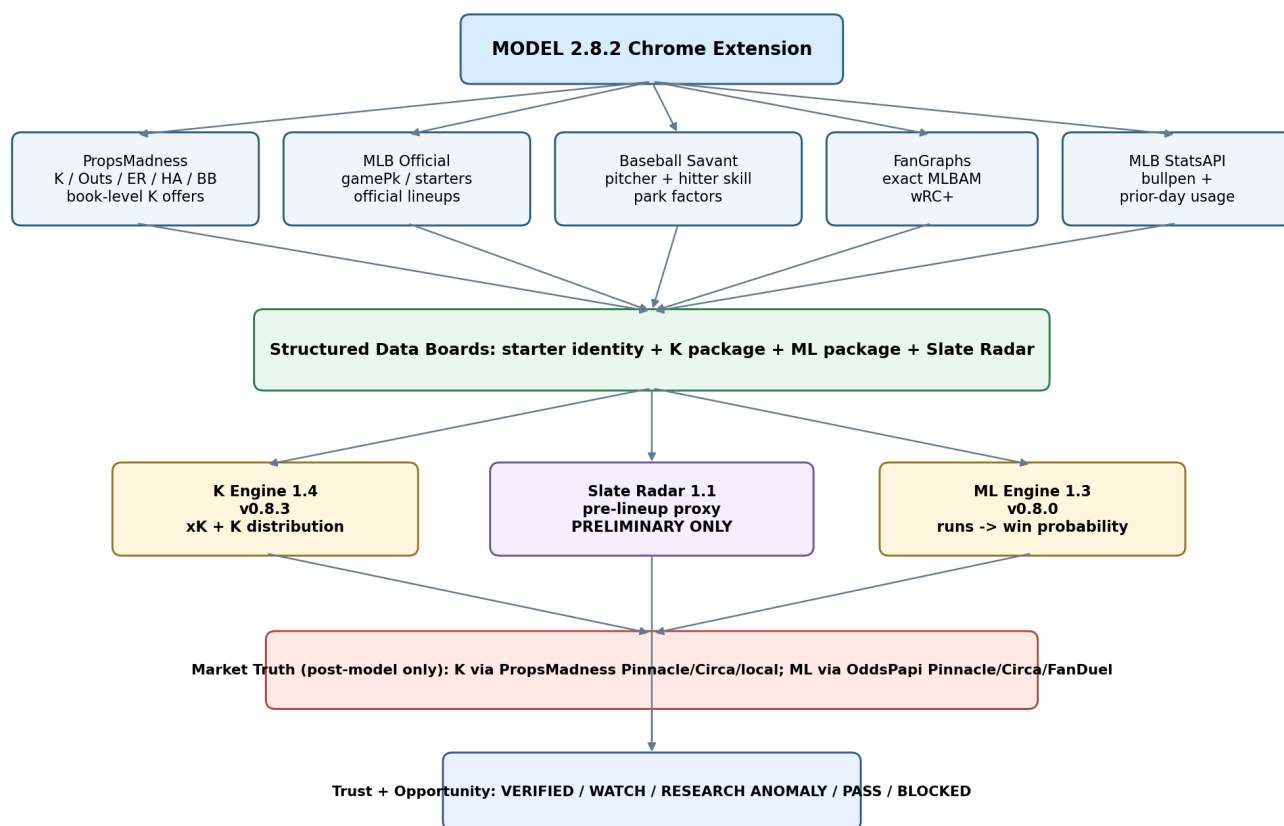
Two-sided no-vig probability:
p_over_fair = p_over_raw / (p_over_raw + p_under_raw)

Edge:
edge = p_model - p_market_fair

Expected value at executable decimal price d:
EV = p_model * d - 1
```

The model can therefore be right about the underlying probability but still have no bet if the price is too expensive. Conversely, a modest probability can be a good wager if the offered price is sufficiently favorable.

4. System Architecture



The architecture has four deliberate layers: source/identity, structured model packages, market truth, and Trust. Slate Radar runs beside production as a non-actionable scouting layer.

```

Chrome Extension
Data Pipeline
  -> PropsMadness table snapshot
  -> MLB Official gamePk / starters / lineups
  -> Savant pitcher + hitter skill
  -> FanGraphs wRC+
  -> MLB StatsAPI bullpen + usage
  -> Savant park factors

Production Boards
  -> K board
  -> ML board

Preliminary Board
  -> Slate Radar (separate storage key; never a Trust input)

Trust Layer
  -> K market truth from PropsMadness
  -> ML market truth from local service / OddsPapi
  -> VERIFIED / WATCH / RESEARCH ANOMALY / PASS / BLOCKED
  
```

Canonical identity. MLB gamePk is the primary event identity. Bookmaker fixture IDs are never treated as universal cross-book identifiers. Each OddsPapi bookmaker event is independently joined to MLB Official, then quotes are merged under gamePk.

5. Constituent Data Sources

Source	Role in MODEL	What it is not
PropsMadness	Pitcher K, Pitcher Outs, Earned Runs, Hits Allowed, Pitcher Walks; recent/season histories; opponent ordinal ranks; book-level K offers.	Not the source of expected-K truth from the current K line; current target K line has zero weight in xK.
MLB Official / StatsAPI schedule	gamePk, teams, game time, official/probable starters, official 9-man batting orders.	Not a performance model. It is the identity authority.
Baseball Savant	Pitcher and hitter K%, Whiff%, SwStr%, Contact%, BB%, batted-ball metrics; Statcast park run factors.	Not a sportsbook price source.
FanGraphs	Exact MLBAM-ID wRC+ for hitters, current season with prior-season handling when appropriate.	Not used as an approximate name join; unresolved exact IDs block production.
MLB StatsAPI bullpen resources	Active roster, season bullpen stats, prior-day usage/rest state.	Not replaced by neutral bullpen values in production when validation fails.
OddsPapi	Supported game-market truth only: Pinnacle, Circa, FanDuel.	Never used for player K props on this plan; never used as a predictive input.

Opponent rankings from PropsMadness

Opponent rankings such as K rank, Whiff rank, Contact rank, BB rank, Chase rank, and xwOBA rank are ordinal 1-30 context only. MODEL intentionally does not convert an ordinal rank into a fabricated percentage. They can be displayed as context but are not allowed to masquerade as raw rates.

Park factors

The park provider prefers a 3-year Baseball Savant run factor. For newer venues, it may fall back to the longest available 2-year or 1-year window. The model clamps the run factor between 0.85 and 1.15. A known example is Sutter Health Park, whose available 2025-2026 run index was 121 and therefore enters the model at the 1.15 clamp.

6. Data Pipeline and Canonical Identity

The Data Pipeline controller is the orchestration layer inside the extension. Its main user-facing function is SYNC + BUILD K + ML. A successful sync builds all current structured boards in one pass and uses zero OddsPapi requests.

SYNC + BUILD K + ML

1. Find an open PropsMadness MLB props-table tab.
2. Initialize the structured table provider in Chrome MAIN world.
3. Capture the five pitcher markets and book-level offers.
4. Fetch MLB Official schedule / probable pitchers / lineups.
5. Build verified starter board keyed by gamePk.
6. Fetch current + previous Savant pitcher/batter data.
7. Fetch FanGraphs wRC+.
8. Fetch bullpen rosters/stats/prior-day usage.
9. Fetch adaptive Savant park factors.
10. Resolve official lineups.
11. Build K production board.
12. Build ML production board.
13. Build Slate Radar.
14. Atomically promote the new current-state storage keys.

Fail-closed promotion. If a sync errors, no new K, ML, or Radar board is promoted. The previous valid current board remains available rather than being replaced by a partially built state.

Production readiness gates

A moneyline game does not become READY simply because a schedule entry exists. Production requires verified starter identity, both official 9-man lineups, complete Savant hitter coverage, exact wRC+ coverage, validated bullpen input, a park factor, and a current-game Pitcher Outs workload anchor for both starters.

K production similarly requires official starter/event identity, the official opponent batting order, structured pitcher/hitter skill, workload/supporting markets where required, freshness, and a usable K market for downstream evaluation.

7. Structured K Model: Exact Calculation Flow

Current lineage. Engine 1.4 - mlb-k-v0.8.3-sample-shrinkage-workload-2026-09-07. Calibration status remains PROSPECTIVE_INPUT_MIGRATION.

The K model has two jobs: estimate the pitcher's expected strikeouts (xK) independently of the target strikeout line, then convert xK and uncertainty into an Over/Under probability at whatever market line is being evaluated.

7.1 Small-sample pitcher skill shrinkage

Current-season pitcher K%, Whiff%, SwStr%, and Contact% are not accepted at full strength until 180 batters faced. If the sample is smaller, each metric is shrunk toward the pitcher's previous season when the prior has at least 60 BF; otherwise it is shrunk toward a league-neutral skill prior.

```
current_weight = clamp(current_BF / 180, 0, 1)

adjusted_metric = current_weight * current_metric
                 + (1 - current_weight) * prior_metric
```

Neutral priors if no credible previous season:

```
K%      22.5
Whiff%   25.0
SwStr%   11.5
Contact% 75.0
```

This rule was added after the Pivetta failure case. A 60-70 BF strikeout spike can still influence the model, but it no longer receives the same authority as a 500-600 BF skill sample.

7.2 Workload state

```
Current Pitcher Outs line:
<= 7.5 outs   -> OPENER_LIKE
<= 14.5 outs  -> LIMITED
> 14.5 outs   -> NORMAL
missing       -> UNANCHORED
```

Limited or opener-like workloads reduce confidence and widen distribution uncertainty. In Trust, unresolved role/workload is a blocking condition.

7.3 Expected workload and batters faced

Supporting PropsMadness markets are used as structural workload/context anchors. This is allowed because they are not the target K market. Market-implied means are combined with recent history and season averages, with neutral fallback values when necessary.

```
Expected ER:   current-market weight .66, recent .34, season component .24
Expected Hits: current-market weight .64, recent .36, season component .24
Expected Walks: current-market weight .64, recent .36, season component .24
Expected Outs: current-market weight .64, recent .36, season component .24
```

```
leash = .28*(ER - 2.6) + .08*(Hits - 5.0) + .15*(BB - 2.0)
leash is clamped to [-0.75, 1.15]
```

```
expected_outs = clamp(base_expected_outs - leash, 3, 23.5)
expected_BF = expected_outs + expected_hits + expected_walks + 0.30
```

The expected BF result is then constrained to a plausible range, with a much lower ceiling for opener-like roles.

7.4 Lineup strikeout matchup

For the official 9-man lineup, hitter K%, Whiff%, SwStr%, and Contact% are plate-appearance shrunk and batting-order weighted. The top of the lineup receives slightly greater weight. The lineup result is also blended with an opponent team prior.

```
opponent_K = 0.72 * lineup_K + 0.28 * team_K
```

```
base_matchup_K = logistic(
    logit(pitcher_K) + logit(opponent_K) - logit(0.225)
)
```

Then adjust the rate for pitcher/opponent Whiff, SwStr and Contact:

```
+ (pitcherWhiff - 25) * .0012
+ (oppWhiff      - 25) * .0009
+ (pitcherSwStr  - 11.5) * .0018
+ (oppSwStr      - 11.5) * .0012
- (pitcherContact - 75) * .0006
- (oppContact    - 75) * .0006
```

matchup_K is clamped to 6% - 46%

```
structural_xK = expected_BF * matchup_K
```

7.5 Structural/recent/season xK blend

Expected K then blends structural xK with recent K history and season K average. The raw weights are structural .55, recent .15, and season .08. Missing components are omitted and the remaining weights are renormalized.

If all three components exist, normalized weights are approximately:

```
structural 70.51%
recent     19.23%
season     10.26%
```

```
xK = weighted mean of available components
physical_ceiling = min(18, expected_BF * 0.75)
xK is clamped to [0.15, physical_ceiling]
```

Target-line separation. The current K line and current K price have exactly zero weight in expected K. They are introduced only after xK exists, when the distribution is evaluated at that line.

7.6 Probability distribution

The engine uses an overdispersed Poisson mixture rather than a single Poisson mean. This widens tails when uncertainty is elevated.

```
uncertainty_multiplier = 1
  + small_sample_uncertainty
  + workload_uncertainty

small_sample_uncertainty = (1 - current_weight) * .35
workload_uncertainty = .25 when workload is limited
multiplier is clamped to 1.0 - 1.7

Let d = .18 * uncertainty_multiplier.
Evaluate five Poisson means around xK:
  xK*(1-d), xK*(1-d/2), xK, xK*(1+d/2), xK*(1+d)
with weights:
  0.08, 0.22, 0.40, 0.22, 0.08

P(Over line) = weighted sum of the five Poisson Over probabilities.
```

8. K Market Comparison and Trust

Once the independent K distribution exists, MODEL evaluates the current PropsMadness K line and compares its model probability with available book-level prices.

Sharp K reference

PropsMadness currently exposes book-level offers. MODEL looks for Pinnacle and Circa/Circa Sports at exactly the same K line. A book contributes only if both Over and Under prices are available, because both sides are required to remove vig.

```
For each sharp book at the target line:
  convert Over and Under American odds to implied probabilities
  no_vig_over = p_over / (p_over + p_under)

If Pinnacle and Circa both qualify:
  sharp_over = average(Pinnacle no-vig Over, Circa no-vig Over)
  sharp_under = 1 - sharp_over
```

If the sharp books quote different K lines, MODEL does not fabricate a consensus. If only one sharp book qualifies, it can be used as a weaker single-sharp reference. A local two-sided price can form a local no-vig reference. A one-sided price can support EV-only WATCH but cannot create a no-vig market probability.

K Trust states

State	Meaning
VERIFIED	High-confidence, fresh production distribution; adequate edge/EV; two-book same-line Pinnacle/Circa sharp consensus; no important warnings. Currently prospective K calibration can downgrade this to WATCH.
WATCH - NOT VERIFIED	Promising but not fully verified: medium confidence, single sharp source, local source, one-sided price, aging market, prospective calibration, or similar caveat.
RESEARCH - ANOMALY	Model/market probability disagreement exceeds the anomaly gate. Research before acting.
PASS	Model works, but edge or EV fails minimum threshold.
BLOCKED	Identity, workload/role, freshness, schema, confidence, market

Current default thresholds include a 12 percentage-point anomaly gate, a 2.5 percentage-point minimum verified edge, and 2% minimum EV. Market snapshots become aging after about 180 seconds and blocked after about 300 seconds. K package freshness is stricter than ML: about 30 minutes maximum source age and 10 minutes maximum capture skew.

Current K calibration policy. K lineage is still prospective after the input migration. Even a mathematically VERIFIED K result is intentionally downgraded to WATCH until the lineage is validated on forward data.

9. Moneyline Model: Exact Calculation Flow

Current lineage. Engine 1.3 - mlb-moneyline-v0.8.0-offense-strength-2026-09-06. The coefficients were frozen during the structured-source migration.

The moneyline model predicts team runs first, then converts the expected run differential into a win probability. Sportsbook moneyline prices do not enter this calculation.

9.1 Lineup offense factor

Each official 9-man lineup is PA-shrunk and batting-order weighted. The offense index uses six metrics around league baselines.

```
offense_index =
  .50 * ((wRC+ - 100) / 25)
+ .18 * ((ISO - .170) / .055)
+ .12 * ((xSLG - .400) / .065)
+ .08 * ((xwOBA - .320) / .040)
+ .07 * ((BB% - 8.5) / 3)
- .05 * ((K% - 22.5) / 5)

index is clamped to [-1.75, 1.75]
raw offense factor = 1 + .085 * offense_index
final factor is clamped to [0.86, 1.14]
```

Production ML requires official lineups, so the official-lineup weight is 1. Radar can use a projected active-roster proxy with a lower lineup weight and a separate uncertainty penalty.

9.2 Starter run-prevention adjustment

starterAdj begins at 1.0

```
Pitcher component adds:
+ (pitcher xwOBA - .320) * 2.10
+ (pitcher BA - .245) * .75
+ (pitcher xSLG - .400) * .45
+ (HardHit% - 39) * .0025
+ (Barrel% - 8) * .0040
- (K% - 22) * .0065
- (Whiff% - 25) * .0035
+ (BB% - 8) * .009
```

Opponent matchup component adds:

```
+ (opp xwOBA - .320) * 1.70
+ (opp BA - .245) * .55
+ (opp xSLG - .400) * .40
+ (opp HardHit - 39) * .0020
+ (opp Barrel - 8) * .0030
- (opp K% - 22) * .0045
+ (opp BB% - 8) * .007
+ (opp Contact - 75) * .0025
```

```
starterAdj = clamp(starterAdj + 0.35 * matchupAdj, 0.60, 1.52)
starter_R9 = 4.35 * starterAdj * offense_factor
```

9.3 Bullpen model

Only validated structured bullpen inputs are used in production. The bullpen model uses robust means so a single extreme relief row does not dominate the entire unit.

```
base bullpen_R9 = .58 * bullpen_ERA + .42 * 4.20

bullpen_R9 *= 1
+ (WHIP - 1.28) * .18
- (K% - 23) * .004
+ (BB% - 8.5) * .006

bullpen_R9 *= offense_factor
bullpen_R9 *= 1 + (1 - freshness) * .05
bullpen_R9 is clamped to [2.7, 6.6]
```

Bullpen freshness is derived from active relievers and prior-day usage. A side is only production-usable when the bullpen validation status is trusted and at least four qualifying relief rows are available.

9.4 Workload, park, runs, and win probability

Production ML requires a current-game Pitcher Outs line. A season-average Outs value is allowed only in Radar/diagnostics. This was hardened after the Chase Burns workload-management case.

```
starter_IP = current Pitcher Outs line / 3

team_runs = starter_R9 * starter_IP / 9
+ bullpen_R9 * (9 - starter_IP) / 9

team_runs *= park_run_factor
if home offense: team_runs += 0.20
team_runs is clamped to [1.8, 7.5]

run_diff = home_runs - away_runs
P(home win) = 1 / (1 + exp(-run_diff / 1.60))
P(away win) = 1 - P(home win)
```

Fair American odds are derived directly from these independent win probabilities. Only after that does OddsPapi enter the process.

9.5 Production ML readiness

- Both starter identities verified.
- Both official 9-man lineups present.
- Nine Savant hitters per opposing lineup.
- Exact FanGraphs wRC+ resolved for all production hitters.
- Validated bullpen for both sides.
- Savant park factor resolved.
- Current-game Pitcher Outs workload anchor for both starters.
- Fresh synchronized model package.

10. Slate Radar: Early Candidate Discovery

Slate Radar exists because official MLB lineups often arrive too close to first pitch. It gives the user time to identify which markets are worth watching before production is ready.

Hard boundary. Radar is stored separately at `model_mlb_slate_radar_current`, has `actionable=false`, uses 0 OddsPapi requests, and is never accepted as a Trust input.

ML Radar

When both starter identities are verified but lineups are not yet official, Radar builds a preliminary offense profile from the MLB active roster. Non-pitchers are enriched with Savant and FanGraphs data, ranked by PA, and the top likely regulars form the proxy. At least seven usable hitters are required; neutral placeholders may fill the remaining proxy slots only for preliminary scouting.

ML Radar uses the same v0.8.0 projection engine, but it carries an uncertainty penalty and a watch-price target. The current watch-price logic reserves approximately a 3 percentage-point model edge before converting the preliminary probability into a maximum acceptable market price.

K Radar

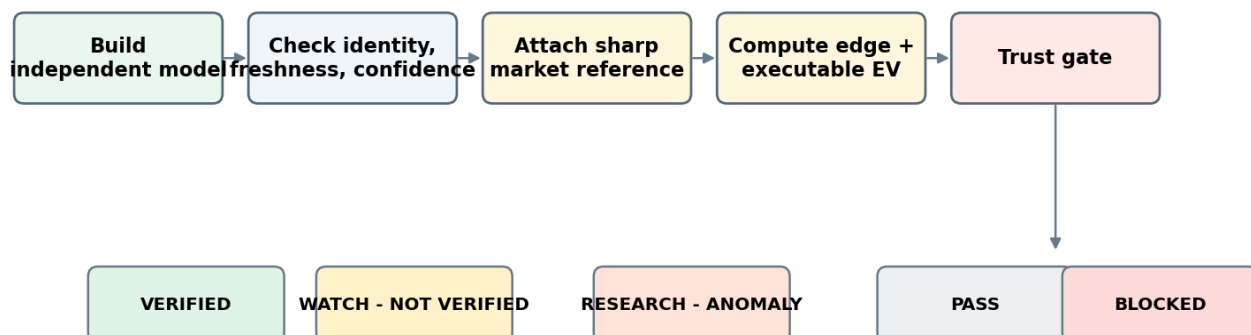
K Radar uses the verified starter, active-roster opponent proxy, pitcher Savant skill, and the K engine. If a K line/price exists, it can show preliminary probability and EV; if no line exists, the distribution can still be shown, but the market state must be NO MARKET and EV remains unavailable.

Radar stages

Stage	Meaning
RADAR	Pre-lineup scouting. Use to identify markets to watch, not to place a final bet.
FINALIZING	One official lineup or other final input is arriving. Rebuild soon.
READY_FOR_MARKET	Production ML board is complete; a game-market refresh can now attach prices.
READY_FOR_K_TRUST	Production K distribution is complete and can be evaluated locally at current K prices.
WAITING_STARTER / WAITING_DATA	Important identity or structured input is unresolved; no probability may be manufactured when the required identity is missing.

11. Trust and Opportunity Integrity

MODEL does not equate a large raw edge with a good bet. Trust can downgrade or block it.



Trust is the risk-control layer. It deliberately has authority to downgrade a mathematically attractive model output when data quality, market quality, confidence, freshness, or disagreement is unacceptable.

Moneyline Trust defaults

```

Market aging warning:    > 180 seconds
Market hard block:      > 300 seconds
Anomaly disagreement:   > 12 percentage points
Minimum verified edge:  >= 2.5 percentage points
Minimum verified EV:    >= 2%
ML input aging warning: > 45 minutes
ML input hard block:    > 90 minutes
ML capture skew hard block: > 30 minutes
  
```

Opportunity score and ordering

MODEL 2.8.2 prevents a huge anomaly EV from outranking a clean WATCH. The score begins with a state base and then adds bounded EV/edge/reference-quality points while subtracting risk penalties.

```

State base:
  VERIFIED 10000
  WATCH    8000
  PASS      2000
  ANOMALY   1000
  BLOCKED    0

Selected risk penalties include:
  Medium K confidence    -60
  Low K confidence       -500
  aging/stale warning    -35
  single sharp book      -35
  single local source    -55
  one-sided K price      -85
  disagreement warning   -75
  prospective calibration -20
  workload/role unresolved -250
  
```

This is a presentation/decision integrity calculation. It does not alter the underlying baseball probability.

12. The Local Market Service

The local service is a small Python service running on the user's Mac at <http://127.0.0.1:8765>. Its purpose is to isolate game-market credentials and market-fetching logic from the Chrome extension, persist timestamped market snapshots, join bookmaker events to official MLB identity, and expose a small authenticated localhost API to the extension.

What the service does

- Reads the OddsPapi credential from the macOS Keychain boundary rather than embedding it in extension code.
- Requires a private per-install localhost token from the extension for authenticated calls.
- Fetches supported MLB game markets from Pinnacle, Circa Sports, and FanDuel.
- Makes one tournament request per bookmaker under the current OddsPapi contract.
- Independently joins each bookmaker fixture to MLB Official and merges quotes under MLB gamePk.
- Computes the downstream sharp game-market reference and best available executable prices.
- Persists immutable timestamped snapshots plus a latest pointer and exposes audit telemetry.

Important endpoints used by the extension

```
/v1/health
  Confirm service identity/version and credential configuration.

/v1/config
  Return Trust/service configuration.

/v1/markets/refresh?sport=mlb
  Perform a supported MLB game-market refresh. Normal policy: about 3 requests.

/v1/audit
  Return latest request log, book coverage, canonical join counts, sharp-ready counts, quota policy.
```

Where it lives

```
Canonical source:
/Users/abbeyfelix/Developer/MODEL/services/market-service/src/model_service.py

Managed deployed copy used by the running service:
~/Library/Application Support/MODEL/service/model_service.py
```

A previous activation bug occurred because the managed Application Support copy remained older than the canonical source. Current installers synchronize the managed deployment when service code changes and verify the authenticated live health version before claiming success.

Quota policy

The OddsPapi plan is currently 250 requests/month and does not include player props. MODEL therefore reserves OddsPapi for supported game markets and deliberately uses 0 OddsPapi requests for K workflows. A normal full MLB moneyline refresh uses approximately 3 requests total for the whole slate - one each for Pinnacle, Circa, and FanDuel.

13. How to Use the Extension

Canonical Chrome source. Chrome should load only /Users/abbeyfelix/Developer/MODEL/apps/chrome-extension/src. Do not load a release ZIP, backup folder, or old extension path.

Normal daily workflow

1. Open the PropsMadness MLB props table in Chrome and allow its pitcher markets to populate.
2. Open MODEL. In the Data Pipeline panel, run SYNC + BUILD K + ML. This uses 0 OddsPapi requests.
3. Review Slate Radar / Opportunity Radar early in the day. Treat it as a shortlist of markets to watch, not a list of bets.
4. As official lineups post, run SYNC + BUILD K + ML again. Rebuilds remain 0 OddsPapi.
5. For K, run REFRESH K EDGES - 0 API. K prices come from PropsMadness, including sharp book-level offers when available.
6. For ML, wait until several games of interest are READY, then run REFRESH ML + ALL - ~3 API once for the remaining slate.
7. Read the Trust panel. Prioritize VERIFIED, then clean WATCH candidates. Research anomalies; ignore PASS/BLOCKED as opportunities.
8. Use COPY TRUST AUDIT when sharing diagnostics or asking for analysis.

Useful buttons

Button	What it does	OddsPapi cost
SYNC + BUILD K + ML	Refreshes structured source data and rebuilds K, ML, and Radar boards.	0
COPY K BOARD	Copies detailed structured K board JSON.	0
COPY ML BOARD AUDIT	Copies detailed production ML board and blockers.	0
COPY DATA AUDIT	Copies full data-pipeline audit.	0
REFRESH K EDGES - 0 API	Evaluates current production K distributions against PropsMadness K market truth.	0
REFRESH ML + ALL - ~3 API	Fetches Pinnacle/Circa/FanDuel game markets and compares production ML probabilities.	~3
COPY TRUST AUDIT	Copies compact user-facing state, market coverage, and ranked Trust results.	0

When NOT to refresh OddsPapi

- When the only change is that a lineup has posted - rebuild first for free.
- When most late games are still BLOCKED.
- When you only want K edges.
- When the local service version is not the expected version.
- When a previous refresh is still fresh enough for the decision you are making.

14. Storage, Persistence, and Credential Boundaries

The extension uses Chrome local storage for current structured boards. Market-service snapshots are persisted separately by the local service. Current-state keys are deliberately version-independent pointers so the latest successful sync can replace the previous board atomically.

Storage key	Purpose
model_pm_table_snapshot_current	Latest structured PropsMadness table snapshot.
model_mlb_official_starter_board_current	Verified schedule/starter identity board keyed by gamePk.
model_mlb_k_projection_board_current	Current production K board.
model_mlb_ml_projection_board_current	Current production moneyline board.
model_mlb_slate_radar_current	Preliminary, non-actionable Radar board.
model_data_pipeline_audit_current	Latest detailed source/build audit.
modelV2LastEdgeBoard	Latest Trust edge board generated by K-only or full market refresh.

Credential boundary

The OddsPapi API key must never be embedded in release payloads or extension source. It lives behind the local service / macOS Keychain boundary. The Chrome extension uses a per-install localhost token to authenticate to the local service. Release installers preserve the installed token and test packages use a placeholder token rather than shipping the real secret.

Archival philosophy

Active code should remain small. When old runtime files are removed, the guarded installer archives them under archive/migrations before deleting them from the active Chrome extension directory. MODEL 2.7.0 physically removed the old PMV1Engine/Hydrate/Batch era from active runtime rather than merely hiding it.

15. Essential Code Modules and Functions

The following names come from the current 2.8.x source and are the most important places to understand when debugging or extending the system.

Module	Essential functions	Responsibility
features/data_pipeline/controller.js	sync, fetchOfficial, fetchSavant, fetchFangraphs, fetchBullpenResources, fetchParkFactorsAdaptive, resolvePregameLineups	Orchestrates the full structured build and promotes current boards.
models/mlb/k/structured_k_core.js	adjustedPitcherSkill, workloadStateFromOuts, buildDistribution, independentBlend, evaluateAtLine, buildBoard	Creates independent xK and the K probability distribution; applies small-sample/workload uncertainty.
models/mlb/moneyline/structured_ml_core.js	teamOffenseStrength, bullpenModel, expectedRunsAgainstStarter, projectGame, workloadFromStarter, buildStructuredBoard	Creates expected team runs and independent game win probabilities.
features/slate_radar/radar_core.js	rosterProfile, officialProfile, mlPreviewForGame, kPreviewsForGame, buildBoard	Builds pre-lineup scouting candidates. Never a Trust input.
model_v2_core.js	trustMoneyline, kSharpReference, localStarterKMarket, trustStarterK, opportunityScore, sortEdgesByOpportunity	Market comparison, no-vig logic, Trust gates, risk penalties, final ordering.
model_v2.js	getModelEdges, refreshKOnly, refreshAndBuild, indexMarketsByGamePk, copyAudit	User-facing Trust panel, local-service client, market attachment, audit generation.
services/market-service/src/model_service.py	service endpoints / book fetch / official join / canonical merge	Secure OddsPapi boundary and canonical game-market snapshot service.

Why mirrored package/core files exist

Some core logic is mirrored between the Chrome extension runtime and packages/core or packages/models paths. The project uses mirror checks so shared logic does not silently diverge. Release validation includes syntax checks, mirror checks, model goldens, source-provider tests, Trust tests, quota-policy tests, and installer simulations.

16. Status Labels: How to Read MODEL

Label	User interpretation
VERIFIED	The highest production state. Required model inputs and market reference passed current integrity rules. Still a probability, not a certainty.
WATCH - NOT VERIFIED	Potentially useful signal with one or more caveats. Investigate price, confidence, source quality, and context before action.
RESEARCH - ANOMALY	The model and market disagree too much. Treat this as a diagnostic/research target, not a stronger bet.
PASS	The market is priced too efficiently for the current model edge/EV threshold.
BLOCKED	MODEL refuses to produce an actionable comparison because a required input, identity, workload, freshness, role, or price condition failed.
RADAR / CANDIDATE	Preliminary early-slate scouting only. Not production and not actionable.
FINALIZING	The game is approaching production readiness; rebuild as remaining official inputs arrive.
NO MARKET	A model distribution may exist, but there is no current betting line/price. EV must remain unavailable.

A BLOCKED row is a feature, not a failure. The system is intentionally designed to prefer no prediction over a polished-looking prediction based on unresolved starter identity, missing official lineups, stale data, unsupported workload, or invalid schema.

17. Troubleshooting and Operating Rules

Service version is old

If Trust shows an older service version than the release expects, do not spend another game-market refresh. The live listener may be running the managed Application Support copy. Use the current guarded installer/repair flow and confirm /v1/health reports the required version before spending quota.

Sharp-ready rows are zero

Use the Trust audit book coverage fields. The 2.3.5 service exposes per-book request outcome, fixtures returned, official joins, and ML quotes parsed. This tells you whether the failure is upstream request coverage, identity join, market parsing, or downstream Trust attachment.

A giant K edge appears

Do not assume it is the best bet. Check K confidence, workload state, sample adjustment, market reference type, freshness, and whether Trust labeled it ANOMALY. The Pivetta case is the canonical example of why raw probability alone is insufficient.

A starter is workload-managed

Production ML now requires the current Pitcher Outs anchor; K labels limited/opening workloads and can block unresolved roles. However, external news context such as a team-announced piggyback or rehab pitch cap is not yet a fully automated provider. If known news conflicts with the current market/workload package, treat the game/prop conservatively and flag it for research.

A Radar name/price looks impossible

Verify the official schedule/starter assignment before assuming the join is wrong, because 2026 roster movement can make familiar players appear on unfamiliar teams. If the current K price is absent, the row should display NO MARKET; 2.8.2 fixed the null-to-zero display bug.

Never do these

- Never use OddsPapi for player props on the current plan.
- Never load Chrome from an old release/backup path.
- Never force a production prediction through an unresolved identity or lineup gate.
- Never average Pinnacle and Circa K probabilities when they are quoting different K lines.
- Never interpret Radar as a final bet recommendation.
- Never refit frozen model coefficients while debugging source plumbing unless the task is explicitly a model-retraining project.

18. Known Limitations and MLB Roadmap

MODEL is now structurally stable, but it is not finished. The most important remaining work is validation and richer context, not another large rewrite of the core equations.

Priority	Current gap	Planned direction
1. Forward validation	K/ML calibration has not yet accumulated a large prospective sample under the current lineages.	Automatic ledger with probability, price, close, result, Brier, log loss, CLV, ROI, edge buckets, model version.
2. Injury/workload news	Current-game Outs and sample uncertainty are modeled, but team-announced pitch caps/piggybacks/IL return context is not yet a fully automated structured provider.	Add explicit workload-event input with credible source, timestamp, confidence and fail-closed rules.
3. Large-edge review band	Anomaly gate is >12pp; an 11.9pp VERIFIED ML edge can still look too similar to a routine 3-5pp edge.	Add REVIEW LARGE EDGE band below anomaly threshold.
4. Opportunity Board polish	Decision hierarchy works but the UI can become a clearer single-screen product.	Unified ranked board with countdowns, stage, confidence, sharp reference, target price, reason decomposition.
5. Explainability	Raw component fields exist but are not fully summarized for the user.	Show why xK / run differential moved: skill, opponent, workload, bullpen, park, lineup.
6. Automated alerts	User still manually rebuilds as lineups post.	Unattended schedule-aware rebuilds and saved/notification state, while minimizing API usage.

Validation metrics that should decide future coefficient changes

- Brier score and reliability/calibration buckets.
- Log loss.
- Closing-line value (CLV), both vs sharp close and executable close when available.
- ROI by edge bucket, price band, favorite/underdog, confidence, workload class, and model version.
- For K: performance by expected BF, sample-size tier, target line, and Over/Under side.

Only after the forward ledger is large enough should MODEL v0.9/v1.0 coefficients be changed. The goal is to learn from evidence rather than tuning to memorable wins/losses.

19. NFL Branch Plan

Status. NFL is the next major branch and is targeted for MODEL 2.9.0. It is not yet a production model in the extension. MLB 2.8.2 is the hardened baseline from which NFL should branch.

The key lesson from MLB is architectural: NFL should begin with canonical identity, structured independent inputs, explicit uncertainty, market truth only after the model, and a Trust layer from day one.

MODEL SPORT CONTROLLER

MLB

-> existing MLB structured pipeline

NFL

-> NFL schedule / canonical game ID
 -> nflverse historical + weekly team efficiency
 -> QB identity/status
 -> injuries / depth chart
 -> weather / rest / venue context where justified
 -> independent game outcome model
 -> OddsPapi game-market truth
 -> NFL Trust / Opportunity Board

Planned NFL feature families

- Passing EPA/play and opponent pass-defense efficiency.
- Rushing EPA/play and opponent run-defense efficiency.
- Success/down efficiency and first-down efficiency.
- CPOE / quarterback passing efficiency.
- Sack/pressure propensity and pass-protection interaction.
- Explosive-play creation/prevention.
- Turnover propensity with heavy regression toward the mean - not raw turnover margin copied as stable skill.
- Red-zone/finishing metrics only if they add out-of-sample value.
- Special teams/kicking where predictive and stable.
- QB identity, continuity, questionable/out status, and injury uncertainty as first-class gates.
- Home field, rest, travel, divisional context, weather when supported by evidence.

NFL modeling standard

The first NFL model should be fit chronologically on historical games, not randomly shuffled. A 2025 holdout should remain untouched during development. The release should report Brier score, log loss, calibration, and comparison against simple baselines. Week 1 should be explicitly prior-heavy and confidence-capped.

PropsMadness NFL may eventually be valuable for player-prop market truth, especially because book-level Pinnacle/Circa prop prices are useful on MLB. However, NFL v0.1 should not depend on an unavailable NFL Table View or a brittle dashboard scraper. An email request to PropsMadness for an NFL Table View/export remains worthwhile.

20. PROJECT CHECKPOINT - Upload This Section to Restore Context

Recovery instruction. If ChatGPT loses project context, upload this PDF and say: "Resume MLB MODEL from the Section 20 checkpoint. Treat it as the authoritative current project state unless I provide newer logs/releases."

20.1 Canonical paths and release discipline

Canonical dev root:

/Users/abbeyfelix/Developer/MODEL

Chrome must load ONLY:

/Users/abbeyfelix/Developer/MODEL/apps/chrome-extension/src

Managed market-service deployment:

~/Library/Application Support/MODEL/service/model_service.py

Release rule:

ZIP + SHA256 + guarded installer + full regression + rollback/archive.

Preserve localhost token. Never ship the real OddsPapi key/token.

20.2 Current production versions

Item	Checkpoint value
Extension	MODEL 2.8.2 - Opportunity Integrity
Data Pipeline	1.9.x structured control plane
Local service	2.3.5
K engine	1.4
K lineage	mlb-k-v0.8.3-sample-shrinkage-workload-2026-09-07
K calibration	PROSPECTIVE_INPUT_MIGRATION
ML engine	1.3
ML lineage	mlb-moneyline-v0.8.0-offense-strength-2026-09-06
ML calibration	PRODUCTION_INPUT_MIGRATION
Slate Radar	1.1 PRELINEUP_ACTIVE_ROSTER_PROXY; preliminary only
Legacy runtime	PMV1Engine/Hydrate/Batch physically removed
OddsPapi policy	250/month; game markets only; K player-prop calls = 0

20.3 Non-negotiable model rules

- Expected K must remain independent of the current target K line/price; TARGET_K_MARKET_WEIGHT = 0.
- PropsMadness supplies K/Outs/ER/Hits Allowed/Walks/history and book-level K prices; K market truth is downstream.
- Pinnacle/Circa K consensus only when both quote the same K line with two-sided prices. Never average different lines.
- OddsPapi is reserved for supported game markets; never build OddsPapi player-prop discovery on the current plan.
- ML probability is built before OddsPapi. Pinnacle/Circa/FanDuel are comparison/pricing inputs only after projection.
- ML production requires official lineups and current-game workload anchors. Season-average workload may be used only by Radar/diagnostics.
- Radar is a separate non-actionable board and must never feed Trust.
- Blocked inputs fail closed. Do not replace unresolved production inputs with neutral defaults.

- Do not refit ML v0.8.0 coefficients during source/plumbing work.

20.4 Current ML math checkpoint

```

Offense index weights:
  wRC+ .50 | ISO .18 | xSLG .12 | xwOBA .08 | BB% .07 | K% -.05
Offense scoring adjustment: .085 * index; factor clamp .86 - 1.14
Starter baseline R9: 4.35; starterAdj clamp .60 - 1.52
Opponent matchup multiplier into starterAdj: .35
Bullpen base: .58*ERA + .42*4.20, then WHIP/K/BB/freshness; clamp 2.7 - 6.6
Park factor clamp: .85 - 1.15
Home run-environment add: +0.20 runs
Team run clamp: 1.8 - 7.5
P(home) = logistic((homeRuns - awayRuns)/1.60)

```

20.5 Current K math checkpoint

```

Small sample threshold: 180 current BF
Neutral priors: K 22.5 | Whiff 25 | SwStr 11.5 | Contact 75
Workload states from current Outs: <=7.5 OPENER_LIKE; <=14.5 LIMITED; else NORMAL
Structural matchup combines pitcher K with official-lineup K plus Whiff/SwStr/Contact adjustments
structural_xK = expectedBF * matchupK
Raw xK blend weights: structural .55 | recent .15 | season .08 (renormalized when available)
Physical ceiling: min(18, expectedBF*0.75)
Over probability: overdispersed five-point Poisson mixture
Current target K line/price: 0 weight in xK

```

20.6 Current market/service checkpoint

- Local service 2.3.5 is live and verified.
- Canonical ML market identity: each bookmaker fixture is independently joined to MLB Official and merged under gamePk.
- Normal full MLB refresh: three book calls - pinnacle, circasports, fanduel.
- Latest live 2.8.2 audit on 2026-09-08 showed 15 sharp-ready rows, 15 two-sharp consensuses, 15 multi-book merged rows, 15 canonical games, player-prop requests 0.
- K book-level PropsMadness offers now preserve Pinnacle/Circa/local quotes; current K calibration normally keeps qualified K plays at WATCH until forward validation.

20.7 Release history that matters

Release	Key milestone
2.6.0	Structured ML production cutover.
2.6.1	OddsPapi one-book request contract fixed; service deployment mismatch diagnosed/repared.
2.6.2	Bookmaker fixture IDs canonicalized under MLB gamePk; sharp consensus live-verified.
2.7.0	Legacy PMV1Engine/Hydrate/Batch runtime physically removed from active extension.
2.8.0	Slate Radar added for early whole-slate candidate discovery.
2.8.1	K sample-size shrinkage + workload uncertainty; ML current-game workload anchor; book-level K sharp references.
2.8.2	Opportunity Integrity: state hierarchy, anomaly quarantine, confidence/risk penalties, no-market null integrity.

20.8 Open issues / next tasks

- Freeze MLB 2.8.2 math while gathering forward evidence. Make only integrity/presentation fixes unless a real defect is found.
- Add a forward-validation/CLV ledger for K and ML: predictions, closing prices, outcomes, Brier, log loss, calibration, ROI, version buckets.
- Add a structured workload/injury-news layer so IL returns, piggybacks, pitch caps, openers, and workload management are machine-readable instead of manual context.
- Consider a large-edge REVIEW band below the 12pp anomaly gate; latest live example LAA@BOS was VERIFIED at an 11.9pp ML edge.
- Continue UI toward a single polished Opportunity Board with stage, confidence, countdown, sharp reference, executable target price, and explainability.
- Begin NFL 2.9.0 from the hardened 2.8.2 baseline. Do not start with a brittle PropsMadness scraper. Use a reproducible nflverse-centered historical pipeline, explicit QB/injury state, chronological training, 2025 holdout, Week-1 uncertainty, downstream OddsPapi game-market Trust.
- Email PropsMadness asking for an NFL Table View/export; MLB book-level Pinnacle/Circa player-prop prices are now valuable downstream inputs.

20.9 Known cautionary cases

- Nick Pivetta: final K model once showed ~6.22 xK / ~73.6% Over 4.5 and he finished with 2 K. This exposed small-sample and return/workload overconfidence; v0.8.3 added shrinkage/uncertainty and Low-confidence blocking.
- Chase Burns: production ML had used season-average workload despite a special curtailed/piggyback plan. v2.8.1 changed production ML to require current-game Pitcher Outs; season average is Radar/diagnostic only.

- Drew Anderson: Radar produced an extreme Under projection; production Trust correctly blocked it on unresolved workload/role. This is the desired Radar -> Trust safety behavior.

20.10 How to resume work safely

1. Confirm installed extension version and local service version from Trust Audit.
2. Confirm Chrome is loaded from the canonical src path only.
3. Read the latest release notes / Trust Audit before changing code.
4. Reproduce the defect with a regression test first.
5. Keep K and ML frozen unless the task is explicitly model retraining.
6. Installer/tests should make 0 OddsPapi requests.
7. Live-market verification should be minimal and quota-aware.
8. Never delete rollback/archive history before a replacement path is live-verified.

End of checkpoint. This checkpoint is intended to be sufficient for a future session to reconstruct the current architecture, version lineages, non-negotiable policies, known failure cases, and the immediate NFL/validation roadmap without relying on conversation memory.